EXPERIMENT – 6

AIM :

To create a stored procedure transfer_employee(emp_id, new_dept_id) with validation and error handling, implement triggers for salary validation and audit logging, and test edge cases.

SOURCE CODE :

```
CREATE TABLE Department (
   Dept_ID INT PRIMARY KEY,
   Dept_Name VARCHAR(50),
   Location VARCHAR(50)
);

CREATE TABLE Project (
   Project_ID INT PRIMARY KEY,
   Project_Name VARCHAR(100),
   Budget DECIMAL(12,2)
);

CREATE TABLE Employee (
   Emp_ID INT PRIMARY KEY,
   Emp_Name VARCHAR(50),
   Salary DECIMAL(10,2),
   Job_Role VARCHAR(50),
   Dept_ID INT,
   Project_ID INT,
   FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID),
   FOREIGN KEY (Project_ID) REFERENCES Project(Project_ID)
);

CREATE TABLE Employee_Audit (
   Audit_ID INT AUTO_INCREMENT PRIMARY KEY,
   Emp_ID INT,
   Old_Salary DECIMAL(10,2),
   New_Salary DECIMAL(10,2),
   Action_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

INSERT INTO Department VALUES
(1, 'Human Resources', 'Delhi'),
(2, 'Information Technology', 'Chandigarh'),
(3, 'Finance', 'Mumbai'),
(4, 'Marketing', 'Bangalore'),
(5, 'Research and Development', 'Hyderabad');

INSERT INTO Project VALUES
(101, 'Website Development', 500000),
```

```
(102, 'Mobile Application', 750000),
(103, 'Cyber Security System', 900000),
(104, 'Data Analytics Platform', 850000),
(105, 'AI Chatbot', 1200000),
(106, 'Cloud Migration', 1000000),
(107, 'Payroll Management', 400000),
(108, 'Market Research System', 600000);

INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 45000, 'HR Executive', 1, 107),
(2, 'Ananya Gupta', 52000, 'HR Manager', 1, 107),
(3, 'Riya Verma', 48000, 'Recruiter', 1, 107),
(4, 'Karan Singh', 55000, 'HR Analyst', 1, 107),
(5, 'Priya Mehta', 60000, 'HR Manager', 1, 107),
(6, 'Rahul Kumar', 75000, 'Software Engineer', 2, 101),
(7, 'Neha Sharma', 82000, 'Software Engineer', 2, 102),
(8, 'Aditya Singh', 95000, 'Security Engineer', 2, 103),
(9, 'Simran Kaur', 88000, 'Data Analyst', 2, 104),
(10, 'Vivek Gupta', 105000, 'Cloud Engineer', 2, 106),
(11, 'Ishita Jain', 78000, 'Software Developer', 2, 101),
(12, 'Rohan Verma', 92000, 'AI Engineer', 2, 105),
(13, 'Aman Yadav', 65000, 'Financial Analyst', 3, 107),
(14, 'Sneha Kapoor', 72000, 'Accountant', 3, 107),
(15, 'Mohit Agarwal', 85000, 'Finance Manager', 3, 107),
(16, 'Pooja Sharma', 68000, 'Financial Analyst', 3, 107),
(17, 'Nikhil Jain', 90000, 'Senior Accountant', 3, 107),
(18, 'Kriti Verma', 62000, 'Accountant', 3, 107),
(19, 'Arjun Malhotra', 58000, 'Marketing Executive', 4, 108),
(20, 'Tanya Singh', 70000, 'Marketing Manager', 4, 108),
(21, 'Yash Gupta', 62000, 'Marketing Analyst', 4, 108),
(22, 'Muskan Sharma', 75000, 'Brand Manager', 4, 108),
(23, 'Dev Kumar', 66000, 'Sales Executive', 4, 108),
(24, 'Naina Kapoor', 80000, 'Marketing Analyst', 4, 108),
(25, 'Kabir Sharma', 95000, 'Research Scientist', 5, 105),
(26, 'Meera Gupta', 88000, 'ML Engineer', 5, 105),
(27, 'Varun Singh', 92000, 'Research Analyst', 5, 104),
(28, 'Aditi Verma', 110000, 'AI Engineer', 5, 105),
(29, 'Harsh Jain', 98000, 'Data Scientist', 5, 104),
(30, 'Isha Kapoor', 85000, 'Research Engineer', 5, 103);

DELIMITER //

CREATE PROCEDURE transfer_employee(
    IN p_emp_id INT,
    IN p_new_dept_id INT
)
BEGIN
    DECLARE v_emp_count INT DEFAULT 0;
    DECLARE v_dept_count INT DEFAULT 0;
```

```
    SELECT COUNT(*)
    INTO v_emp_count
    FROM Employee
    WHERE Emp_ID = p_emp_id;

    IF v_emp_count = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee does not exist';
    END IF;

    SELECT COUNT(*)
    INTO v_dept_count
    FROM Department
    WHERE Dept_ID = p_new_dept_id;

    IF v_dept_count = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Department does not exist';
    END IF;

    UPDATE Employee
    SET Dept_ID = p_new_dept_id
    WHERE Emp_ID = p_emp_id;
END //

CREATE TRIGGER validate_salary_before_insert
BEFORE INSERT ON Employee
FOR EACH ROW
BEGIN
    IF NEW.Salary <= 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be greater than zero';
    END IF;
END //

CREATE TRIGGER validate_salary_before_update
BEFORE UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF NEW.Salary <= 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be greater than zero';
    END IF;
END //

CREATE TRIGGER salary_audit_after_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF OLD.Salary <> NEW.Salary THEN
```

```
        INSERT INTO Employee_Audit
        (Emp_ID, Old_Salary, New_Salary)
        VALUES
        (OLD.Emp_ID, OLD.Salary, NEW.Salary);
    END IF;
END //

DELIMITER ;

CALL transfer_employee(1, 2);

SELECT Emp_ID, Emp_Name, Dept_ID
FROM Employee
WHERE Emp_ID = 1;

UPDATE Employee
SET Salary = 50000
WHERE Emp_ID = 1;

SELECT * FROM Employee_Audit;

CALL transfer_employee(999, 2);

CALL transfer_employee(1, 999);

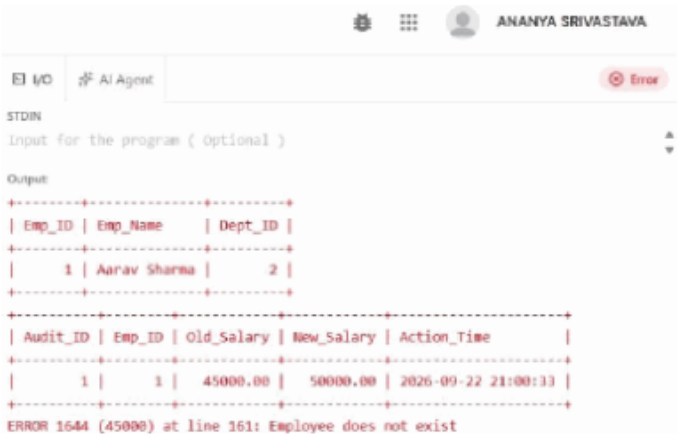
UPDATE Employee
SET Salary = -5000
WHERE Emp_ID = 1;
```

OUTPUT :

Employee Transfer & Salary Audit



Invalid Employee ID



Invalid Department ID



Invalid Salary

